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PERFORMANCE SPECIFICATION

EXPEDITIONARY SHELTER PROTECTION SYSTEM (ESPS) PANEL MODULE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Record of Revisions

REV	CHANGED BY	REVISED PAGES	DESCRIPTION	DATE	APPROVAL
A	R. Devine	3.5,8,11, 15	Drawings update, requirements update, verification update	1/27/2016	R. Devine
B	R. Devine	4, 7	Requirements update	9/28/2016	R. Devine
C	N. Kakalecz	3 through 26	Drawings updates, figure updates, requirements updates per Developmental Testing (DT), verification update based on data collected during DT and new resin performance.	6/20/2019	N. Kakalecz

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1. SCOPE

1.1. Scope. This specification covers the performance, safety, and environmental requirements of one panel module, be it a single panel or a combination of panels, of the Expeditionary Shelter Protection System (ESPS). The ESPS is comprised of the panel module (the armor material) and a variety of auxiliary structural support equipment, as indicated by the listing of drawings provided below. However, this specification pertains to the requirements and verifications for the ESPS panel module only.

1.2. Classification. The Expeditionary Shelter Protection System (ESPS) is a modular armor system for protection of shelters, equipment, supplies, and temporary fighting positions.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see section 6.2).

Aircraft Materials Fire Test Handbook DOT/FAA/AR-00/12 – Vertical Bunsen Burner Test For Cabin and Cargo Compartment Materials

(This specification is available at the Federal Aviation Administration William J. Hughes Technical Center's Full-Text Technical Reports page: www.actlibrary.tc.faa.gov in Adobe Acrobat portable document format (PDF))

SAE-AMS-STD-595 – Colors Used in Government Procurement

MIL-STD-662 – Department of Defense Test Method Standard, V_{50} Ballistic Test for Armor

MIL-STD-810 – Test Method Standard for Environmental Engineering and Laboratory Tests

MIL-STD-1472 – Design Criteria Standard Human Engineering

(Copies of these documents are available online at https://quicksearch.dla.mil)

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2.2.2. Other government documents, drawings, and publications.

Dwg 5-10-2020 – EXPEDITIONARY SHELTER PROTECTION SYSTEM (ESPS)

Dwg 5-10-2021 – SUPPORT STRUT, WELDED ASSEMBLY, ESPS

Dwg 5-10-2022 – UNIVERSAL STRUT HANDLE, ESPS

Dwg 5-10-2023 – BLAST PANEL, ESPS

Dwg 5-10-0208 – GROUND ANCHOR BRACKET, ESPS

(Copies of drawings are available from the US Army Research Development and Engineering Command, Natick Soldier Research Development and Engineering Center, ATTN: AMSRD-NSR-ST-P, Natick, MA 01760-5018)

2.3. Non-government documents.

ASTM-D570 – Standard Test Method for Water Absorption of Plastics

ASTM-D7264/D7264M-21 – Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials

ASTM-D4812 – Standard Test Method for Unnotched Cantilever Beam Impact Resistance of Plastics

AATCC-119 – Color Change Due to Flat Abrasion (Frosting): Screen Wire Method

(Copies of these documents are available online at https://www.astm.org)

AATCC-169 – Weather Resistance of Textiles: Xenon Lamp Exposure

(A copy of this document is available online at https://www.aatcc.org)

2.4. Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified (see section 6.2), samples shall be subjected to first article inspection (see section 6.3) in accordance with section 4.2.

3.2. Construction. Materials and construction methods for the ESPS panel module are the choice and responsibility of the contractor and shall be capable of meeting all of the requirements specified herein.

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3.3. Workmanship. The ESPS panel module, be it a single panel or combination of panels, shall conform to the quality of product established by this performance specification. Utmost care shall be taken during fabrication to ensure quality workmanship and safety of the service person using the item. Manufacturing practices shall be capable of consistently yielding product that conforms to all requirements in this specification and any internal specifications. Continual improvement shall also be a constant focus of the manufacturing practices. All component materials shall be properly marked and identified, and also protected properly during storage. Materials shall be produced and integrated to extend durability and provide consistency of appearance throughout system's life. All components to be assembled shall be thoroughly cleaned of all foreign matter. In any case of material, process, or equipment change desired to be made by the contractor, the written consent of the Government shall be obtained before making the change. Additional testing may be required prior to implementation of the change to verify product performance.

3.4. Operational requirements. The ESPS panel module, be it a single panel or combination of panels, shall satisfy the following user-oriented requirements. The ESPS panel module shall be capable of being assembled and disassembled in the field with no special tools (if the drawings provided above are complied with, this capability will be confirmed). Additionally, the ESPS panel module shall be capable of being assembled and disassembled with no more than four (4) personnel (if the weight requirement below is complied with, this capability will be confirmed).

3.4.1. Dimensions. ESPS panels shall conform to the dimensions shown in NSRDEC drawing 5-10-2023. All dimensions are in inches. The length and width of the ESPS panel have an allowable tolerance of ± 0.25 inches. The thickness of any individual ESPS panel shall not exceed 0.75 inches and shall not vary over the total area by more than 0.15 inches (uniformity). Cut out locations have an allowable tolerance of ± 0.125 inches and cut out dimensions have an allowable tolerance of ± 0.0625 inches including any edging materials. The cut out areas shall not impede the use of the support strut or the ground anchor bracket. See section 4.6.1 for verification procedure.

3.4.1.1. Panel edging. ESPS panels shall have a firmly bonded or panel-integral edging that protects the perimeter of the panel and any cut outs from liquid absorption, delamination, cuts, wear, and tear. The edging itself shall also be resistant to cutting, wear, and tear. The edging may extend in from the sides (overlap) of the panel a maximum of 0.50 inches. If the edging is non-integral to the panel, then it shall consist of a one-piece or continuously bonded construction. The edging shall be firmly and completely attached to the panel, i.e. there shall be no unbonded areas. If over-molded edging is utilized, ensure that the panel face to edging transition has a minimum radius of 0.0625-inch. The edging shall be smooth as to not cause or potentially cause harm to the user. The edging shall match the color of the panel itself or be black. See section 4.6.1 for verification procedure.

3.4.2. Weight. ESPS strike panels shall each weigh less than 5.36 pounds per square foot. The ESPS panel module (comprised of a combination of backer and strike panels) shall weigh less than ten and a half (10.61) pounds per square foot total. See section 4.6.2 for verification procedure.

3.4.3. Labeling. The ESPS panel module shall be labeled in exact accordance with NSRDEC drawing 5-10-2023. See section 4.6.3 for verification procedure.

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3.4.4. Color. ESPS panels faces shall be painted using CARC paint. The panels shall be Tan on one side and Green on the other. Special requests can be made for panels painted white on both faces.

a. Tan 686A (33446)
b. Green (383)
c. White 506 (27885)

Colors shall conform to SAE-AMS-STD-595 – Colors Used in Government Procurement. See section 4.6.4 for verification procedure.

3.4.5. Health and safety. The ESPS panel module, be it a single panel or combination of panels, shall be non-hazardous to the immediate user and personnel in the surrounding area.

3.4.5.1. Flame resistance. The average flame time for each material type tested, vertical and horizontal, shall be as stated in TABLE I. There shall be no melted or flaming drips. The average burn length for each material type, vertical and horizontal, shall be as specified in TABLE I. See section 4.6.5.1 for verification procedure.

TABLE I. **Flame Resistance Requirements**

	12 sec Impingement	60 sec Impingement
Vertical Flame Time	NTE 2.0 sec	NTE 60.0 sec
Vertical Burn Length	NTE 2.0 inches	NTE 2.0 inches
Horizontal Flame Time	N/A	NTE 2.0 sec
Horizontal Burn Diameter	N/A	NTE 4.0 inches
* Please note that each of these values will be an average of the three specimens evaluated.		

3.5. Ownership and support. The ESPS panel module, be it a single panel or combination of panels, shall satisfy the following ownership and support requirements.

3.5.1. Edge impact resistance. ESPS panels shall have their edging strength assessed to ensure the edging withstands typical use in its operating environment without deterioration. Panel edge strength shall meet or exceed 3.70 ft·lb/in when tested on an IZOD impact machine in accordance with ASTM D 4812 “Standard Test Method for Unnotched Cantilever Beam Impact Resistance of Plastics.” See section 4.7.1 for verification procedure.

3.5.2. Abrasion. Abrasion. ESPS panels shall have their abrasion resistance tested to ensure the panels withstand typical use in their operating environment without deterioration. The panel specimen (4 inch diameter) shall not experience any signs of cracking, flaking, chipping or separation of the coating from the substrate in accordance with AATCC 119 - Color Change Due to Flat Abrasion (Frosting): Screen Wire Method. Results of the test shall be expressed as "pass" or "fail" as exhibiting visible coating nonconformities. See section 4.7.2 for verification procedure.

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3.5.3. Delamination. ESPS panels shall show no signs of delamination (separation into constituent layers) upon visual inspection or acoustic interrogation. See section 4.7.3 for verification procedure.

3.5.4. Field agent resistance. ESPS panels shall show no evidence of softening, peeling, delamination, ply separation, or tackiness when exposed or subjected to the following agents:

a. DEET insect repellent, NSN 6840-01-284-3982, O-I-503 Type II, Concentration A
b. Gasoline, ASTM D4814
c. Motor Oil, MIL-PRF-2104
d. Hydraulic fluid, petroleum base, MIL-PRF-6083
e. Fire resistant hydraulic fluid, MIL-PRF-46170
f. Fuel Oil, Diesel, ASTM D975
g. Turbine Fuel, Aviation, JP-8, MIL-DTL-83133
h. Rifle Bore Cleaning Compound, NSN 6850-00-224-6656, MIL-PRF-372
i. Lubricating Oil, Weapons (LSA), NSN 9150-00-935-6597
j. Lubricating Oil, Arctic, Weapons, NSN 9150-00-292-9689
k. Face paint, NSN 6850-01-493-7309
l. Aviation Gasoline, ASTM D910, Grade 100LL

See section 4.7.4 for verification procedure.

3.6. Operating environment requirements. The ESPS panel module, be it a single panel or combination of panels, shall withstand both common and extreme environmental conditions and shall not degrade in any way when exposed to those conditions.

3.6.1. Weatherometer resistance. ESPS panels shall exhibit no evidence of cracking, blistering, edge separation, delamination, or otherwise degradation after being subjected to Weatherometer resistance testing. See section 4.8.1 for verification procedure.

3.6.2. Ultraviolet (UV) light resistance. ESPS panels shall exhibit no evidence of cracking, blistering, edge separation, delamination, or otherwise degradation after being subjected to prolonged ultraviolet (UV) light exposure. See section 4.8.2 for verification procedure.

3.6.3. Water absorption. ESPS panels shall not increase in weight by more than 0.5 percent during twenty-four (24) hour fresh water immersion or increase in weight by more than 1.0 percent during long-term fresh water immersion. If moisture absorption does occur, the panel shall remain within the weight requirements of section 3.4.2. See section 4.8.3 for verification procedure.

3.6.4. Hot storage. ESPS panels shall exhibit no evidence of cracking, blistering, edge separation, delamination, or otherwise degradation after being subjected to hot conditioning. Additionally, thickness measurements are to be performed at six (6) intersecting locations on the panel before and after conditioning per FIGURE 1. Each of these six thickness measurements, when compared before and after conditioning, shall not vary by more than 10% of the original thickness. See section 4.8.4 for verification procedure.

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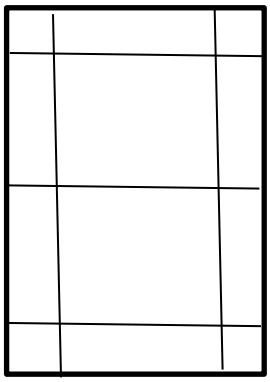


FIGURE 1. **Measurement Locations**
Note that each Line is ~2.5" (+/- 1/16") from panel edge and horizontal line is mid panel.

3.6.5. Cold storage. ESPS panels shall exhibit no evidence of cracking, blistering, edge separation, delamination, or otherwise degradation after being subjected to cold conditioning. Additionally, thickness measurements are to be performed at six (6) intersecting locations on the panel before and after conditioning per FIGURE 1. . Each of these six thickness measurements, when compared before and after conditioning, shall not vary by more than 10% of the original thickness. See section 4.8.5 for verification procedure.

3.7. Performance requirements. The ESPS panel module, be it a single panel or combination of panels, shall provide ballistic protection from indirect (fragments) as well as direct (firearm) fire.

3.7.1. FAT Ballistic performance. The ESPS panel module, be it a single panel or combination of panels, shall meet the ballistic requirements set forth in TABLE II at 0-degree obliquity with the specified projectiles under the following conditions:

a. Ambient temperature
b. At -25°F
c. At 130°F

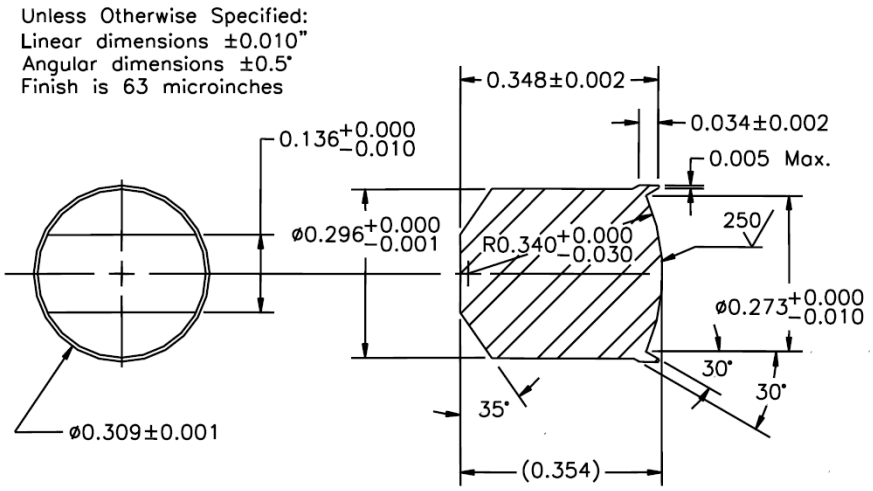
TABLE II. FAT **Ballistic requirements**

Projectile	Minimum V_{50} BL(P) at 0° Obliquity (ft/sec)	V_0 Resistance to Penetration (ft/sec)
7.62×39mm API BZ	2,500	2,350 – 2,400
7.62×54mm LPS Ball	2,700	2,550 – 2,600
7.62×51mm M80 Ball	2,850	2,700 – 2,750

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44-grain FSP 1/	4,275	N/A

1/ Fragment-Simulating Projectile shall be in accordance with Figure 2 below (MIL-DTL-46593B with Amendment 1, 1 April 2006)



Notes:

1. Adjust length to meet the desired weight (grains).
2. Material is AISAI 4340 heat treated to Rockwell "C" hardness of 30 ±1.

FIGURE 2. **Fragment-simulating projectile**

This protection level shall be provided over the entire ESPS panel module except for the panel edges, which shall comply with section 3.4.1.1 above. See section 4.9.1 for verification procedure and details.

3.7.2. LAT Ballistic performance. The ESPS panel module, be it a single panel or combination of panels, shall meet the ballistic requirements set forth in TABLE III at 0-degree obliquity with the specified projectiles under an ambient temperature condition.

TABLE III. LAT **Ballistic requirements**

Projectile	Minimum V_{50} BL(P) at 0° Obliquity (ft/sec)	V_0 Resistance to Penetration (ft/sec)
7.62×39mm API BZ	NA	NA
7.62×54mm LPS Ball	NA	NA
7.62×51mm M80 Ball	NA	2,700 – 2,750
44-grain FSP 1/	4,275	N/A

1/ Fragment-Simulating Projectile shall be in accordance with Figure 2 below (MIL-DTL-46593B with Amendment 1, 1 April 2006)

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3.7.3. Flexural characteristics. The ESPS panel module, be it a single panel or combination of panels, shall withstand blast overpressures from relevant threats and remain in place to perform the system's objective of providing ballistic protection. ESPS panels or any other system components shall not become projectiles or cause any harm to personnel in a blast event. Therefore, the ESPS panel module, be it a single panel or combination of panels, shall meet the flexural characteristics in Table IV below. These characteristics shall be determined using a three-point bend test, in accordance with ASTM D7264 "Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials." In addition, the panel specimen shall exhibit no delamination during or following the test. See section 4.9.2 for verification procedure.

TABLE IV. **Flexural characteristics**

Flexural Modulus (Young's)	> 3,300 ksi
Flexural Stress at Yield	> 14,850 psi

4. VERIFICATION

4.1. Verification alternatives. Alternative test methods, techniques or equipment including the application of cost effective sampling procedures may be proposed by the contractor. Acceptable alternative verification approaches shall be identified in the contract or amended into the contract.

4.2. Classification of testing. The testing requirements specified herein are categorized as First Article Testing (FAT) and Lot Acceptance Testing (LAT) as specified in TABLE's II, III, IV and V.

4.2.1. First article testing (FAT). Unless otherwise specified in the contract or purchase order, the ESPS panel module supplier is responsible for all FATs and LATs herein. The government reserves the right to perform any of the tests set forth in this specification where such tests are deemed necessary to ensure supplies and services conform to prescribed requirements. When a FAT is required, it includes all of the verifications listed in sections 4.6, 4.7, 4.8, and 4.9 (see Table IV) unless otherwise specified in the contract.

4.2.2. Lot acceptance testing (LAT). LAT of ESPS panel modules shall include those verifications listed in sections 4.6, 4.7, 4.8, and 4.9 (see Table V) as defined in the contract or by the procuring activity (see paragraph 6.2). LAT shall occur at a frequency of once per 500 units produced, unless otherwise specified by the Government.

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4.2.3. First Article Testing (FAT) and Lot Acceptance Testing (LAT).

TABLE V. **Requirements and verifications**

CHARACTERISTICS	REQUIREMENT SECTION	VERIFICATION SECTION	FIRST ARTICLE TESTING (FAT)	LOT ACCEPT-ANCE TESTING (LAT)
Operating Requirements	**3.4**	**4.6**		
Dimensions	3.4.1	4.6.1	X	X*
Panel Edging	3.4.1.1	4.6.1	X	X*
Weight	3.4.2	4.6.2	X	X*
Labeling	3.4.3	4.6.3	X	X*
Color	3.4.4	4.6.4	X	CoC
Health and Safety	3.4.5	4.6.5		
Fire Resistance	3.4.5.1	4.6.5.1-2	X	CoC
Ownership and Support	**3.5**	**4.7**		
Edge Impact Resistance	3.5.1	4.7.1	X	CoC
Abrasion	3.5.2	4.7.2	X	CoC
Delamination	3.5.3	4.7.3	X	X*
Field Agent Resistance	3.5.4	4.7.4	X	CoC
Operating Environment Requirements	**3.6**	**4.8**		
Weatherometer Resistance	3.6.1	4.8.1	X	NA
Ultraviolet (UV) Light Resistance	3.6.2	4.8.2	X	NA
Water Absorption	3.6.3	4.8.3	X	NA
Hot Storage	3.6.4	4.8.4	X	NA
Cold Storage	3.6.5	4.8.5	X	NA
Performance Requirements	**3.7**	**4.9**		
Ballistic Performance	3.7.1-2	4.9.1		
V_{50} Ballistic Limit	3.7.1-2	4.9.1.1	X	X**
V_0 Resistance to Penetration	3.7.1-2	4.9.1.2	X	X**
Flexural Strength	3.7.3	4.9.2	X	NA

Notes:

1. X - testing required
2. CoC - certificate of conformance to include data
3. X* - test or inspection to be performed at vendor facility
4. X** - the government may elect to perform testing at their discretion
5. NA - not applicable

4.3. Order of testing. Performing the various testing (operational, ownership & support, operating environment, and performance) can occur in any order.

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4.4. Verification methods. The types of verification methods included in this section are visual inspection, dimensional measurements, sample tests, component properties analysis, and similarity to previously approved or previously qualified designs.

4.4.1. Verification using standard samples. Use standard samples to verify colors with visual inspections.

4.5. Responsibility for compliance. Production items shall meet all requirements specified in section 3. The supplier shall establish and maintain documented procedures for inspection and testing activities in order to verify that the specified requirements for the product are met. The required inspection, testing and the records to be established shall be detailed in a quality plan available to the government as specified in the contract or procuring activity. The inspection set forth in this specification shall become part of the contractor's overall inspection procedures or quality system. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of the manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.6. Operational requirements. All verifications in this section shall be completed.

4.6.1. Dimensions. ESPS panels shall be visually inspected and subsequently measured using certified and calibrated devices to the nearest 0.01 inches, dimensions in excess of 6 inches will utilize a calibrated tape with a +/- tolerance of 1/16". ESPS panels shall conform to the dimensions and tolerances stated in NSRDEC drawing 5-10-2023 and section 3.4.1 of this specification.

4.6.2. Weight. ESPS panels shall be weighed on a certified and calibrated scale to the nearest 0.1 pounds. Total panel weight (lbs) shall be divided by the total surface area of the panel ($28ft^2$) to calculate its areal density (pounds per square foot (psf)). ESPS panels shall conform to all requirements specified in section 3.4.2.

4.6.3. Labeling. ESPS panels shall be visually inspected for legible, permanent labels in conformance with NSRDEC drawing 5-10-2023 as specified in section 3.4.3.

4.6.4. Color. ESPS panels shall be visually inspected for conformance with the standard sample colors specified in section 3.4.4.

4.6.5. Health and safety. All verifications in this section shall be completed.

4.6.5.1. Flame resistance. ESPS panel modules shall undergo vertical and horizontal flame resistance testing in accordance with the "Aircraft Materials Fire Test Handbook DOT/FAA/AR-00/12, Chapters 1 and 3 respectively, with modifications as specified in 4.6.5.1.1 and 4.6.5.1.2. Specimens may be conditioned at $70^{o} \pm 5^{o}$F and 65% $\pm$ 5% relative humidity for 24 hours minimum. Each specimen type shall conform to requirements specified in section 3.4.5.1.

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4.6.5.1.1. Vertical flame resistance. For each material type, six (6) 3-inch x 12-inch specimens shall undergo flame resistance testing. Three of these specimens shall be exposed to a 12-second impingement flame and three shall be exposed to a 60-second impingement flame. Specimens shall be cut with edging reflecting that of the finished ESPS panel module on the 3-inch edge exposed to the flame. In addition, all material types shall be layered and tested as a system, 3 specimens each for 12-second and 60-second tests. Data shall be reported individually and averaged for each material type. Results will be reported and conform as indicated in section 3.4.5.1, TABLE I.

4.6.5.1.2. Horizontal flame resistance. For each material type, three (3) 6-inch x 6-inch specimens shall be prepared with a 1-inch diameter hole cut into the center of the specimen, through all layers of the specimen, and shall undergo 60-second flame resistance testing. During the test, the specimen shall be positioned with the specimen surface perpendicular to the direction of the flame source, with the bottom surface ¾-inch above the burner orifice, and with one edge of the hole centered over the flame source. Each material type shall be tested individually (3 specimens each) and all material types shall also be layered and tested as a system (3 specimens) with the “inside” or ground-facing side of the system facing the flame. Burn rate shall not be calculated. Burn length shall be measured as the diameter of the area of the specimen with evidence of damage due to that area’s combustion including areas of partial consumption, charring, embrittlement, or delamination but not including areas sooted or stained. Data shall be reported individually and averaged for each material type.

4.7. Ownership and support. All verifications in this section shall be completed.

4.7.1. Edge impact resistance. ESPS panels shall undergo edge impact testing on an IZOD impact machine (or a similar device otherwise approved in the test standard) in accordance with ASTM D4812 “Standard Test Method for Unnotched Cantilever Beam Impact Resistance of Plastics.” Testing shall be conducted to the test standard with the point of impact occurring at the interaction line between the panel and edging. ESPS panels shall meet or exceed the energy requirement specified in section 3.5.1.

4.7.2. Abrasion. ESPS panels shall undergo abrasion testing in accordance with AATCC 119 - Color Change Due to Flat Abrasion (Frosting): Screen Wire Method. Section 7. Procedure will be adjusted for non-textile use: The specimen will be placed directly on the reciprocating table, no specimen holder shall be used. Apply head weights in the range of 1134g-3402g. Once the 1200 cycles have completed, the specimen shall undergo visual inspection. The specimen shall not experience any signs of cracking, flaking, chipping or separation of the coating from the substrate as specified in section 3.5.2.

4.7.3. Delamination. ESPS panels shall be visually inspected and tested for delamination using acoustic interrogation using a tap hammer. The full surface of the panel shall be interrogated in a 6-inch x 6-inch array pattern. A bright, metallic sound indicates good structure. A dull "thud" indicates a flaw such as a delamination or disbond. No such delamination shall be evident from either visual inspection or acoustic interrogation as specified in section 3.5.3.

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4.7.4. Field agent resistance. ESPS panels shall be tested for field agent resistance at standard ambient conditions in accordance with the following procedure:

1. With a clean cloth remove any mold release, dirt, or foreign matter from the exterior of the panel.
2. Mark an area of at least 6 inches square on the panel's surface.
3. Apply a liberal amount of the agent specified to the panel sufficient to cover the total marked area for a minimum period of 24 hours. The area shall remain wet with the agent for a minimum of 24 hours. If the agent appears to be close to drying out during the test period, the agent shall be reapplied. It may be useful to use an absorbent pad saturated with the agent to maintain the wetting of the panel.
4. At the end of the test period, remove any excess agent from the panel by dabbing with a dry cloth.
5. Visually examine the test specimen for conformance with the requirements specified in section 3.5.4. Failure to meet any requirement shall constitute test failure.

4.8. Operating environment requirements. All verifications in this section shall be completed.

4.8.1. Weatherometer resistance. ESPS panels shall be tested for weather resistance in accordance with AATCC 169. Specimen size shall be 6-inch x 6-inch. Upon completion of testing, the panel specimen shall be visually inspected for conformance with the requirements specified in section 3.6.1.

4.8.2. Ultraviolet light damage. Accelerated weathering procedure (xenon exposure) AATCC Method 169 shall be utilized except that the following deviations shall apply:

a. The test apparatus shall be either test chamber type 1A or IB. Type 1B shall be equipped with a three-tiered inclined specimen rack. The apparatus shall be equipped with an automatic light monitor and shall be capable of automatically controlling irradiance, temperature, and humidity. The apparatus shall be maintained in accordance with the manufacturer's recommendations.

b. The weathering test cycle shall be 40 minutes of light, 20 minutes of light with water spray on the fabric face, 60 minutes of light, 60 minutes of darkness. This test cycle shall be repeated for a total time period of 200 hours.

c. The irradiance level shall be 0.55 (+/-) 0.01 watt/square meter/nanometer (w/m2/nm) bandpass at 340 nanometers.

d. The glass filter combination shall be a quartz inner filter and a borosilicate type "S" outer filter.

e. The relative humidity shall be 50 (+/-) 5 percent during the light cycle and not lower than 95 percent during the dark cycle.

f. The control set points shall be as follows:

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	Dark cycle Light cycle	Light Cycle
Black panel	38°C	77°C
Conditioning water	40°C	53°C
Wet bulb depression	0°C	10°C

Upon completion of testing, the panel specimen shall be visually inspected for conformance with the requirements specified in section 3.6.2. Discoloration may occur which should be noted but does not signify a failure.

4.8.3. Water absorption. ESPS panels shall undergo water absorption testing in accordance with ASTM D570 "Standard Test Method for Water Absorption of Plastics." The testing procedures to be used are 7.1, Twenty-Four Hour Immersion and 7.4, Long-Term Immersion. A number of specifications stated in the standard are impractical for this product, so the following modifications should be made: (1) the test specimen for this testing shall be one full-size ESPS panel of each panel variety that comprises an ESPS module (i.e. if an ESPS module is comprised of two different panel types, each type shall be tested for water absorption); (2) for each procedure, the specimens shall be immersed in fresh water, not distilled water, at ambient/room temperature, not a precisely conditioned temperature; and (3) only the percent increase in weight of one specimen, not three, is required. Before and after testing, the panels shall be weighed on a certified and calibrated scale to the nearest 0.1 pounds so that their percent increase in weight can be determined. ESPS panels shall conform to the requirements specified in section 3.6.3.

4.8.4. Hot storage. ESPS panels shall be subjected to a conditioning of 24 hours (± 1 hour) at 160°F (± 10°F) in a circulating air test chamber. The test specimen shall then be allowed to return to ambient temperature. The test specimen shall then be removed from the test chamber and visually examined for conformance with the requirements specified in section 3.6.4. Additionally, thickness measurements are to be performed with a certified and calibrated device at six (6) locations per FIGURE 1 on the panel before and after conditioning. These thickness values shall conform to the requirements specified in section 3.6.4.

4.8.5. Cold storage. ESPS panels shall be subjected to a conditioning of 24 hours (± 1 hour) at minus 60°F (± 10°F) in a test chamber. The test specimen shall then be allowed to return to ambient temperature. The test specimen shall then be removed from the test chamber and visually examined for conformance with the requirements specified in section 3.6.5. Additionally, thickness measurements are to be performed with a certified and calibrated device at six (6) locations per FIGURE 1 on the panel before and after conditioning. These thickness values shall conform to the requirements specified in section 3.6.5.

4.9. Performance requirements. All verifications in this section shall be completed.

4.9.1. Ballistic performance. The ESPS panel module, be it a single panel or combination of panels, shall be subjected to V_{50} ballistic limit testing and V_0 resistance to penetration testing to verify its performance against indirect (fragments) as well as direct (firearm) fire.

4.9.1.1. V_{50} ballistic limit. The ESPS panel module, be it a single panel or combination of panels, shall be subjected to V_{50} ballistic limit testing in accordance with MIL-STD-662 to validate the module's

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performance against the projectiles specified in section 3.7.1 at the temperature conditions specified in section 3.7.1. V_{50} shall be calculated as follows:

> *V_{50} BL(P) calculation:* The arithmetic mean of three (3) Partial Penetration (PP's) and three (3) Complete Penetrations (CP's) within a 125 ft/sec velocity spread or four (4) Partial Penetration (PP's) and four (4) Complete Penetrations (CP's) within a 150 ft/sec velocity spread yield the minimum allowable V_{50} BL determination that will be accepted as reliable test results. If six (6) partial penetrations occur that are above the minimum V_{50}, the system demonstrates a V_{50} above the minimum, the system passes, and that V_{50} test is terminated. All shots' velocities are recorded.

Shot locations will be marked by the contactor. Test samples shall be cut according to FIGURE 3 below, with each shot placed at least 2.5 inches from any edge. Note: not all ten (10) shots may be required to adequately calculate the V_{50} BL(P). Each panel module's calculated V_{50} ballistic limits shall meet or exceed the requirements specified in section 3.7.1.

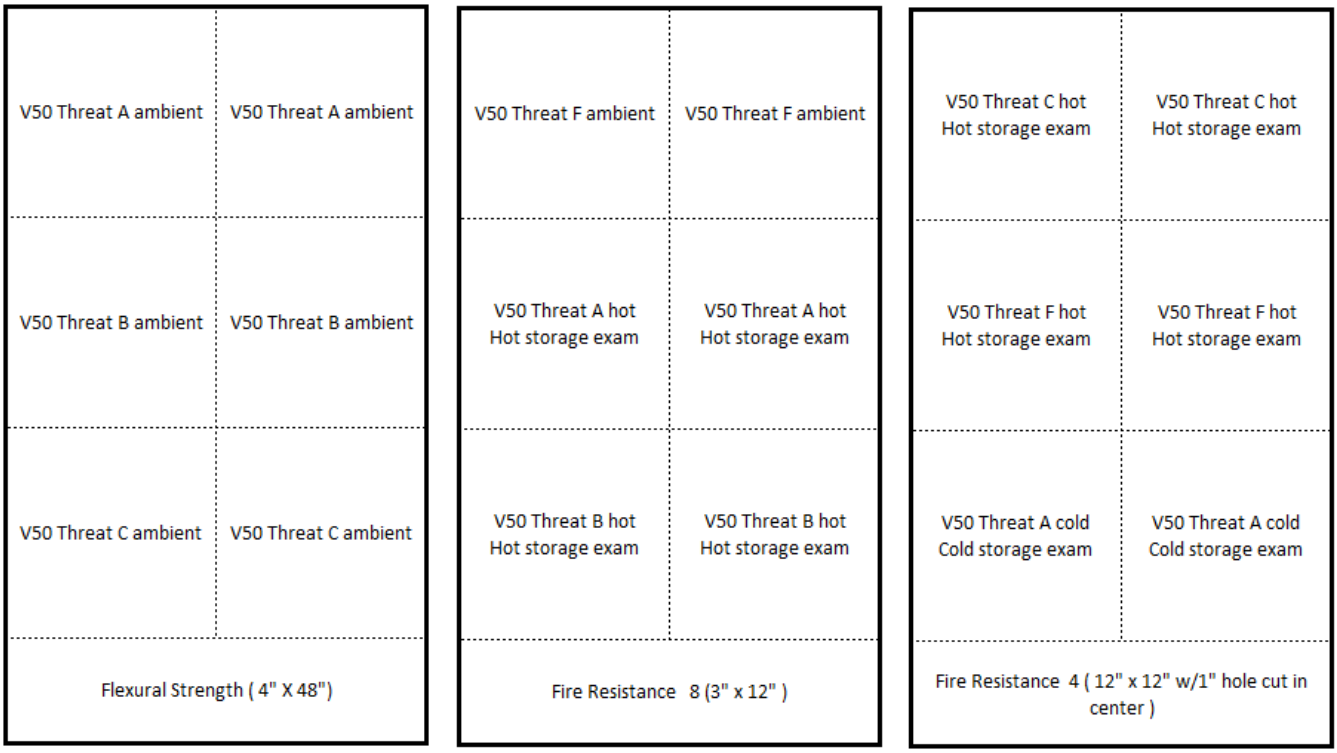


FIGURE 3. **ESPS FAT Test Sample Cut Pattern (Panels 1-3)**

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V50 Threat B cold Cold storage exam	V50 Threat B cold Cold storage exam	V0 Threat A ambient	V0 Threat A ambient	Extra Ballistic	Extra Ballistic
V50 Threat C cold Cold storage exam	V50 Threat C cold Cold storage exam	V0 Threat B ambient	V0 Threat B ambient	Water Absorption & Delamination	Water Absorption & Delamination
V50 Threat F cold Cold storage exam	V50 Threat F cold Cold storage exam	V0 Threat C ambient	V0 Threat C ambient	Abrasion (4" diameter circles) Edge impact Resistance	Weatheromter Resistance / UV Resistance -9 (6" x 6")
Field Agent Resistance 1-2 (1 x 2)	Field Agent Resistance 3- 4 (1 x 2)	Field Agent Resistance 5-6 (1 x 2)	Field Agent Resistance 7-8 (1 x 2)	Field Agent Resistance 9-10 (1 x 2)	Field Agent Resistance 11-12 (1 x 2)

FIGURE 3. **ESPS FAT Test Sample Cut Pattern (Panels 4-6)**

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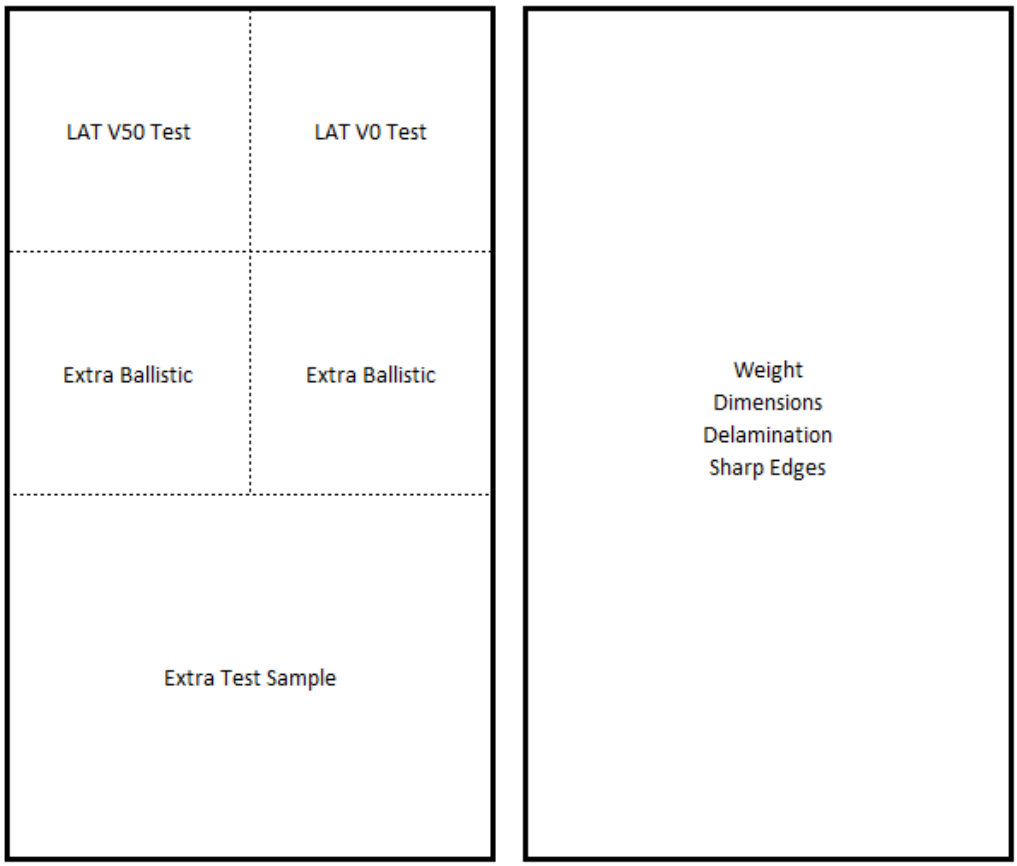


FIGURE 3. **ESPS LAT Test Sample Cut Pattern**

4.9.1.1. V_0 resistance to penetration. The ESPS panel module, be it a single panel or combination of panels, shall be subjected to V_0 resistance to penetration testing to validate the module's performance against the projectiles specified in section 3.7.1 at the temperature conditions specified in section 3.7.1. The V_0 testing of ESPS panel modules shall include a minimum of two (2) fair impacts at 0-degree obliquity: an "Edge" fair impact and a "Face" fair impact. An "Edge" fair impact shall be between 2.0 to 3.0 inches from any edge of armor material and at least 6.0 inches from any previous impact that may have occurred on the panel module. A "Face" fair impact shall be 5.0 – 6.0 inches from any edge impact location and at least 6.0 inches from any previous impact that may have occurred on the panel module. The Government reserves the right to adjust shot spacing to target perceived weak points in the armor panel. The panel module shall suffer zero Complete Penetrations (CPs) throughout V_0 resistance to penetration testing to pass. Complete penetration is defined in NIJ 0106.01 as passage of light thru the witness plate.

Yaw

a.) The yaw card shall be made of stiff material, devoid of any markings and placed directly in front of the test sample. Kodak photographic paper, single weight, kodabromide (or equal), may be used for the yaw card. If photographic paper is used, the emulsion side shall face the test sample.

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b.) After the test, the yaw card shall be carefully measured to determine the largest dimension of the projectile. An optical magnification device with a magnification between 5X and 10X may be used for making this measurement.

c.) In the event that any shot fired indicates yaw greater than 5 degree, it shall not be used and the barrel shall be rechecked for effect on yaw as follows: fire five (5) shots, if three (3) shots exhibit yaw, discard the barrel, and if one or more shots exhibit yaw, investigate cause and correct.

d.) In case of dispute concerning a particular barrel, yaw shall be measured by a photographic measurement system using a multi-flash light source to determine projectile velocity and yaw.

4.9.2. Flexural characteristics. The ESPS panel module, be it a single panel or combination of panels, shall be subjected to a three-point bend test in accordance with ASTM D7264 “Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials.” The specimen size for this test shall be four (4) inches wide, with a length 20% longer than the support span used in the test apparatus. The support span shall be determined using the recommended 32:1 span-to-thickness ratio. The specimen thickness shall be in accordance with the dimensional requirements set in section 3.4.1. The ESPS panel module shall conform to the requirements specified in section 3.7.2.

5. PACKAGING

5.9. Packaging. For acquisition purposes, the contract or order specifies packaging requirements (see section 6.2). When DoD personnel perform material packaging, those personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. The Inventory Control Point packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command, maintains packaging requirements. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.9. Intended use. The ESPS is intended for use in military base camps to provide ballistic protection from direct and indirect fire for shelters and their inhabitants, equipment, supplies, and/or temporary fighting positions.

6.10. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and amendments
b. Type and class (see section 1.2)
c. Issue of DODISS to be cited in the solicitation (see section 2.2.1)
d. When first article inspection is required (see sections 3.1, 4.2, and 6.3)
e. Packaging requirements (see section 5.1)

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6.11. First article. When requiring a first article inspection, contracting documents will provide specific guidance to offerors. This guidance will cover whether the first article is a first article sample, a first production item, or a number of items to be tested. These documents will also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products, which wish to rely on such production or test, should furnish evidence with the bid, that prior Government approval is appropriate for the pending contract.

6.12. Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of the factors include: contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from Table V and their designated frequency based on a risk assessment for the procurement.

6.13. Standard color samples. Standard Color Samples are available from the contracting officer. Class I and Class II colors are similar to Camouflage Green 483 and Tan 686A in SAE-AMS-STD-595.

6.14. Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

6.15. Definitions.

Fair Impact. A fair impact is an impact of an unyawed projectile of the specified projectile type, velocity, and obliquity (±3 degrees), which adheres to shot spacing and edge distance constraints. Any impact which violates shot spacing or edge distance is not considered a fair impact.

Complete Penetration (CP) for V50 Testing. A complete penetration occurs when the impacting projectile or any fragment thereof, or any fragment of the test specimen perforates the rear surface of the armor system.

Complete Penetration (CP) for V0 Testing. A complete penetration occurs when the impacting projectile, any fragment of the projectile, or any fragment of the test specimen fully penetrates the witness plate identified by light transmission through the witness plate..

Partial Penetration (PP). Any fair impact that is not a complete penetration is considered a partial penetration.

Areal Density (AD). A measure of the weight of the armor per unit area, usually expressed in pounds per square foot (lb/ft^2) or kilograms per square meter (kg/m^2) of surface area.

Obliquity. The angle, in degrees, between the line-of-flight of the bullet and a line perpendicular to the front surface (facing the line-of-fire) of the armor system. A projectile's line-of-flight which is

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perpendicular to the surface of the armor system strikes at zero (0) degrees of obliquity.

Yaw. Projectile yaw is the angular deviation of the longitudinal axis of the projectile from the line of flight at a point as close to the impact point on the target as is practical to measure.

V_{50} Ballistic Limit (BL). In general, the velocity at which the probability of penetration of an armor material is 50 percent.

6.16. Subject term (key word) listing.

Armor
Ballistic

6.17. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL:

Custodians:
Army – GL

Preparing Activity:
Army – GL
(Project 5450-2025-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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